Arithmetic Series

Exam code: 9MA0

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Arithmetic series

How do I find the sum of an arithmetic series?

- An **arithmetic series** is the sum of the terms of an **arithmetic sequence**

- The following formulae will let you find the sum of the first ***n*** terms of an arithmetic series:

$$S_n = \frac{n}{2}(2a + (n-1)d) \text{ or } S_n = \frac{n}{2}(a + l)$$

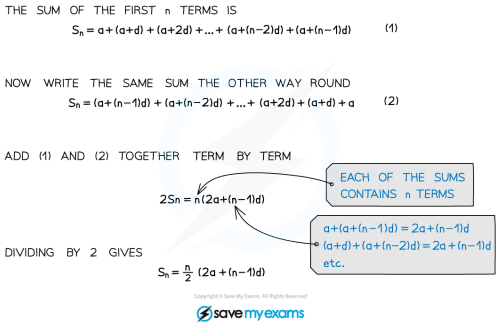
  - *a* is the first term
  - *d* is the common difference
  - *l* is the last term

- You can use whichever formula is more convenient for a given question
- The ***a*** and the ***d*** in those formulae are exactly the same as the ones used with **arithmetic sequences**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Learn this proof of the arithmetic series formula – you can be asked to give it on the exam:
  - Write the terms out once in order
  - Write the terms out again in reverse order
  - Add the two sums together
    - The terms will pair up to give the same sum $2a + (n-1)d$
    - There will be *n* of these terms
  - Divide by two as two of the sums have been added together



Examiner Tips and Tricks

The arithmetic series formulae are in the formulae booklet – you don't need to memorise them.

Worked Example

The fifteenth term is 46. The sum of the first n terms is 2500.

a) Show that $3n^2 + 5n - 5000 = 0$.

b) Hence find the value of n.

a)

$a+(9-1)d = a+8d = 28$ ①

$a+(15-1)d = a+14d = 46$ ②

SIMULTANEOUS EQUATIONS

$6d = 18$

$d = 3$

② MINUS ①

$a = 28 - 8(3)$

$a = 4$

SUBSTITUTE INTO ①

SO $\frac{n}{2}(2(4)+(n-1)(3)) = 2500$

S_n FORMULA

$n(3n+5) = 5000$

$3n^2 + 5n - 5000 = 0$

b)

$(3n+125)(n-40) = 0$

$n = -\frac{125}{3}$ OR $n = 40$

SOLVE THE QUADRATIC — YOU COULD ALSO USE THE QUADRATIC FORMULA OR YOUR CALCULATOR

BUT n MUST BE A POSITIVE INTEGER

$n = 40$

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